ShieldSG: An Attention-Head Suppression Framework for Robust Toxicity Moderation under Adversarial Attacks

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Abstract*—Online toxicity moderation systems must remain reliable even when users deliberately modify harmful text to bypass automated filters using adversarial word substitutions. Although transformer-based classifiers such as BERT achieve high accuracy on standard benchmark datasets, their internal attention mechanisms often contain hidden vulnerabilities that can be exploited by adaptive attackers. This paper addresses the problem of improving the robustness of transformer-based toxicity classifiers under realistic adversarial conditions while preserving clean prediction performance. To address this challenge, we introduce ShieldSG, the novel framework developed in this paper for strategically robust toxicity moderation through attention-head analysis, selective suppression, adversarial evaluation, and Stackelberg game-theoretic defense modeling. The framework first identifies crucial attention heads supporting clean classification and vulnerable heads responsible for adversarial misclassification, then evaluates suppression strategies under adaptive attacker behavior and selects an equilibrium defense policy using a leader–follower optimization formulation. Experimental evaluation on the Jigsaw Toxic Comment dataset shows strong in-domain performance with F1-score of 0.8404 and accuracy of 96.76%, while cross-domain testing on the ToxiGen dataset reveals a significant robustness gap with F1-score of 0.1768. Under adversarial conditions, BERT-Attack achieves a 68.4% attack success rate, but selective suppression of the top 30 vulnerable attention heads improves adversarial accuracy by 28.95 percentage points with minimal loss in clean accuracy. The Stackelberg equilibrium defense further achieves 57.46% adversarial accuracy with 99.85% clean accuracy, and complete adversarial fine-tuning increases robustness to 89.4%. These results demonstrate that ShieldSG provides an interpretable and effective framework for strategically robust transformer-based content moderation.

***Index Terms*—Toxicity Moderation, Adversarial Attacks, Attention Head Suppression.**

I. Introduction

Online platforms increasingly rely on automated toxicity detection systems to moderate harmful content at scale. Transformer-based language models such as BERT have become the dominant backbone for modern moderation pipelines due to their strong contextual understanding and high classification accuracy on benchmark datasets. However, recent studies have shown that these models remain vulnerable to adversarial word-substitution attacks in which attackers modify only a few tokens while preserving the original sentence meaning. As a result, high benchmark accuracy alone is not sufficient to guarantee reliable deployment in adversarial environments where users actively attempt to bypass moderation systems.

A key challenge in building robust moderation systems is that adversarial vulnerability often originates inside the internal attention mechanisms of transformer architectures rather than at the input level alone. Standard robustness evaluations typically measure only attack success rates without identifying which internal components of the model contribute to adversarial failure. This makes it difficult to design targeted defenses that improve robustness without degrading clean prediction performance. In practical moderation settings, an effective defense must therefore preserve normal classification accuracy while limiting the attacker's ability to exploit hidden model pathways.

To address this challenge, this paper introduces ShieldSG, the novel framework developed in this paper for strategically robust toxicity moderation through attention-head analysis, selective suppression, adversarial evaluation, and structured attacker–defender interaction modeling. The framework combines mechanistic interpretability with adversarial robustness analysis by identifying crucial attention heads that support clean classification and vulnerable attention heads that contribute disproportionately to adversarial misclassification. These measurements are then used to construct structured suppression strategies and evaluate their effectiveness under adaptive attacker behavior. Finally, the framework models moderation as a sequential attacker–defender interaction and selects an optimal suppression policy that anticipates attacker best responses.

Experiments are conducted using the Jigsaw Toxic Comment Classification dataset as the primary supervised training benchmark and the ToxiGen dataset as an out-of-domain robustness evaluation dataset. The baseline classifier achieves strong in-domain performance with an accuracy of 96.76% on the Jigsaw dataset, but performance drops significantly on ToxiGen with an F1-score of 0.1768, highlighting the importance of robustness-oriented analysis. Under adversarial evaluation using BERT-Attack, the classifier experiences a 68.4% attack success rate, demonstrating the effectiveness of semantically preserved token-level attacks against transformer moderation systems and motivating the need for structured

TABLE I
FEATURE COMPARISON OF EXISTING ROBUSTNESS APPROACHES AND THE SHIELDSG FRAMEWORK DEVELOPED IN THIS PAPER.

Feature	**Adversarial Attack Literature**	**Attention-Head Analysis Literature**	**Robust Moderation Datasets**	**ShieldSG Framework**
Word-substitution adversarial evaluation	✓	×	×	✓
Attention-head importance mapping	×	✓	×	✓
Vulnerable-head suppression strategy	×	×	×	✓
Adaptive attacker–defender simulation	×	×	×	✓
Cross-domain robustness evaluation (Jigsaw $\rightarrow$ ToxiGen)	×	×	✓	✓
Mechanistic interpretation of adversarial pathways	×	Partial	×	✓
Structured sequential defense strategy selection	×	×	×	✓
Integrated robustness evaluation pipeline	×	×	×	✓

robustness-aware defense strategies.

The primary contributions of this paper are as follows:

- We introduce ShieldSG, a novel robustness framework that integrates attention-head interpretability, adversarial vulnerability detection, selective suppression, and sequential attacker–defender strategy selection for transformer-based toxicity moderation.
- We identify crucial attention heads supporting clean classification and vulnerable attention heads exploited during adversarial word-substitution attacks, and show that selective suppression of the top 30 vulnerable heads improves adversarial robustness by 28.95 percentage points with minimal reduction in clean accuracy.
- We demonstrate through adaptive multi-round attacker–defender simulation that reactive suppression strategies collapse under repeated adversarial adaptation, motivating the need for forward-looking defense selection strategies.
- We validate an optimized suppression strategy that preserves near-baseline clean accuracy while significantly improving robustness, and further strengthen the model through adversarial fine-tuning to achieve 89.4% adversarial accuracy under strong attack conditions.

II. RELATED WORK

Research in adversarial NLP and toxic content moderation spans multiple areas including transformer-based toxicity classification, adversarial attack generation, interpretability analysis, and robustness optimization. Although these studies provide important insights into adversarial robustness and transformer behavior, most existing approaches treat adversarial defense, interpretability, and adaptive robustness as separate problems. Very limited work integrates attention-head vulnerability analysis, adaptive adversarial evaluation, and strategic defense optimization into a unified moderation framework. Table I summarizes the key feature distinctions between existing robustness approaches and the proposed ShieldSG framework.

A. Transformer-Based Toxicity Moderation Models

Recent advances in toxic content moderation have been strongly influenced by transformer-based language models. Devlin et al. [12] introduced BERT, which significantly improved contextual language understanding and became the foundation for many toxicity classification systems. Subsequent transformer architectures such as RoBERTa [13] and DeBERTa [14] further improved representation learning and downstream classification accuracy.

Large-scale toxicity datasets also played an important role in advancing moderation research. Wulczyn et al. [1] introduced the Jigsaw Toxic Comment Classification dataset for large-scale toxic comment detection, while Vidgen et al. [2] developed ToxiGen to evaluate implicit and adversarial hate speech detection. Although transformer-based moderation systems achieve strong benchmark performance, they remain highly vulnerable to adversarial manipulation and cross-domain generalization failures. As shown in Table I, robust moderation datasets primarily address cross-domain generalization but do not provide adversarial-pathway analysis or strategic defense modeling.

B. Adversarial Attack Methods in NLP

Adversarial machine learning has become a major research area for evaluating neural network robustness. Goodfellow et al. [3] first introduced adversarial examples in deep neural networks, followed by Madry et al. [4], who proposed adversarial training and robust optimization techniques. Carlini and Wagner [5] later demonstrated that many existing defenses fail against adaptive attacks.

In NLP, adversarial attacks evolved from character-level perturbations to semantically meaningful word-substitution attacks. Ebrahimi et al. [7] introduced HotFlip for gradient-based adversarial token manipulation, while Alzantot et al. [8] proposed population-based semantic adversarial attacks. Ren et al. [9] further improved semantic attack generation through probability-weighted word saliency techniques. Jin et al. [10] and Li et al. [11] demonstrated that transformer models such as BERT are highly vulnerable to contextual word-substitution attacks capable of preserving semantic similarity while bypassing classifiers effectively.

Although these attacks achieve high attack success rates, most prior studies focus primarily on attack generation rather than analyzing why transformer vulnerabilities emerge internally. As reflected in Table I, adversarial attack literature contributes word-substitution adversarial evaluation but lacks attention-head importance mapping, vulnerable-head suppression, and adaptive attacker–defender simulation.

C. Transformer Interpretability and Attention Analysis

Several studies have investigated the internal behavior of transformer attention mechanisms. Michel et al. [15] analyzed the importance of transformer attention heads and showed that many heads can be removed with limited performance loss. Voita et al. [16] demonstrated that specific attention heads encode specialized linguistic functions, while Clark et al. [17] explored attention behavior inside BERT models.

Tenney et al. [18] showed that BERT implicitly reconstructs the classical NLP pipeline across transformer layers. Hewitt and Manning [19] introduced structural probing methods for analyzing linguistic representations inside neural networks. Rogers et al. [20] summarized major findings in transformer interpretability research through the field commonly referred to as "BERTology."

These works provide valuable insights into transformer behavior, but they do not specifically analyze adversarially vulnerable attention pathways or their role in toxic content moderation robustness. As Table I indicates, the attention-head analysis literature offers only partial mechanistic interpretation of adversarial pathways and does not extend toward suppression-based defense or strategic robustness optimization.

D. Adversarial Robustness and Defense Strategies

Recent research has increasingly focused on improving robustness of NLP systems under adversarial conditions. Vorobeychik and Kantarcioglu [6] provided a comprehensive overview of adversarial machine learning techniques and defense strategies. Morris et al. [24] introduced TextAttack, a framework for adversarial attacks, adversarial training, and robustness evaluation in NLP systems.

Furniturewala and Zubiaga [31] proposed attention-head suppression for adversarially robust content moderation, demonstrating that selective suppression of transformer heads can improve robustness under attack conditions. Sanh et al. [21] further explored selective parameter pruning through movement pruning techniques, showing that transformer representations can be selectively removed while preserving overall performance.

However, most existing robustness approaches rely on static defense strategies and do not model adaptive attacker-defender interactions under dynamically changing adversarial environments.

E. Adaptive Robustness and Strategic Defense Modeling

Recent studies have emphasized the growing need for adaptive and strategically robust moderation systems. Zhang et al. [22] explored equilibrium-based optimization in sequential multi-agent decision-making systems. Tao et al. [23] analyzed robustness limitations of large language models under adversarial attacks, while Himmi et al. [25] discussed challenges associated with robustness benchmarking and evaluation consistency.

Research on modern content moderation systems has also expanded toward large-scale and multimodal moderation challenges. AlDahoul et al. [26] evaluated large language models for detecting sensitive content across multiple modalities, while Chen et al. [27] reviewed robustness, scalability, and explainability challenges in LLM-based moderation systems. Rendo [28] additionally highlighted risks associated with unreliable or excessive automated moderation systems.

Despite these advancements, most existing approaches treat adversarial robustness, interpretability, and adaptive defense optimization independently. They typically rely on static robustness evaluation and do not integrate attention-head vulnerability analysis with adaptive defense modeling and strategic optimization. As summarized in Table I, ShieldSG is the only framework that simultaneously incorporates word-substitution adversarial evaluation, attention-head importance mapping, vulnerable-head suppression, adaptive attacker–defender simulation, cross-domain robustness evaluation, mechanistic interpretation, structured sequential defense selection, and an integrated robustness evaluation pipeline.

The framework proposed in this paper addresses this gap by combining adversarial attack generation, transformer attention-head interpretability, vulnerable-head suppression, adaptive attacker-defender simulation, probing analysis, and game-theoretic robustness optimization into a unified and strategically robust toxicity moderation framework.

III. Methodology

A. Overview of the Proposed Framework

The proposed framework, ShieldSG, is designed to improve the robustness of transformer-based toxicity moderation systems against adversarial attacks while preserving high clean classification performance. As illustrated in Fig. 1, the framework combines adversarial attack generation, attention-head interpretability analysis, selective suppression, adaptive robustness evaluation, and game-theoretic defense optimization into a unified moderation pipeline.

Initially, a baseline BERT toxicity classifier is trained and evaluated on both in-domain and cross-domain datasets to identify generalization weaknesses and adversarial vulnerabilities. Attention-head importance analysis is then performed to identify crucial and vulnerable transformer heads responsible for clean classification and adversarial misclassification.

Adversarial examples are generated using semantic and embedding-based attack methods, after which selective suppression strategies are applied to reduce exploitable transformer pathways while minimizing clean accuracy degradation. Additional probing and circuit-level analyses are conducted to study the linguistic and structural behavior of vulnerable attention heads.

The framework further incorporates adaptive attacker-defender simulations and a game-theoretic optimization model

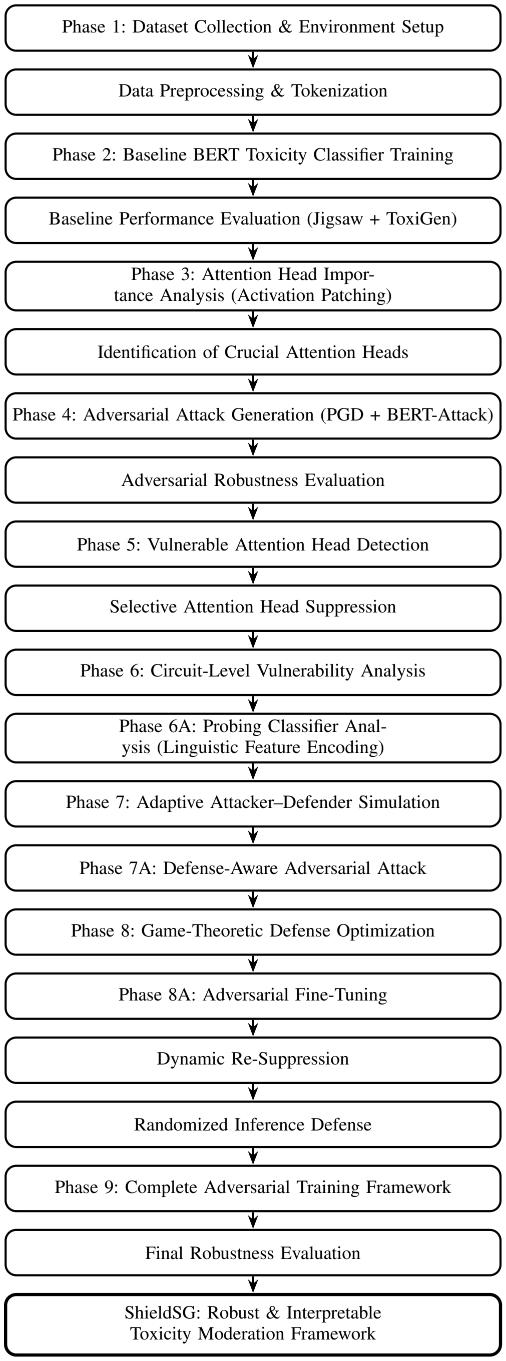

Fig. 1. End-to-end workflow of the proposed ShieldSG framework, illustrating the sequential progression from baseline classifier construction through adversarial analysis, vulnerability-driven suppression, adaptive simulation, game-theoretic optimization, and final adversarial hardening.

to determine the optimal defense strategy under realistic adversarial conditions. Finally, adversarial fine-tuning, dynamic re-suppression, and randomized inference are integrated to further strengthen robustness against adaptive attacks. The complete sequence of phases comprising the framework is depicted in Fig. 1.

Overall, ShieldSG provides an interpretable and strategically optimized framework for robust toxicity moderation under adversarial environments.

B. *Phase 1: Environment Setup and Dataset Preparation*

The first phase focused on establishing a fully reproducible experimental environment using Kaggle Notebooks equipped with NVIDIA Tesla P100 GPUs (16 GB VRAM). The implementation was built using PyTorch and HuggingFace Transformers 5.2.0. To ensure reproducibility across all experiments, a fixed random seed of 42 was applied to every dataset split and training procedure.

Two datasets were incorporated into the framework:

- Jigsaw Toxic Comments Dataset [29] (∼160,000 samples) as the primary training dataset for toxicity classification.
- ToxiGen Annotated Dataset [30] (9,900 samples) for cross-dataset generalization and adversarial robustness evaluation.

Both datasets were split using a stratified 70/15/15 train-validation-test strategy to preserve class balance. A maximum sequence length of 128 tokens was selected since it covered approximately 95% of all Jigsaw comments while remaining computationally feasible on the available GPU resources.

All trained models, adversarial samples, checkpoints, and experimental outputs were stored as persistent Kaggle datasets to ensure reproducibility and continuity across all phases of the research framework.

C. *Phase 2: Baseline Toxicity Classification using BERT*

A baseline toxicity classifier was developed by fine-tuning the BERT-base-uncased transformer model on the Jigsaw dataset. The model architecture consisted of:

- 12 transformer encoder layers
- 12 attention heads per layer
- 768 hidden dimensions
- Approximately 110 million trainable parameters

The output corresponding to the [CLS] token was passed through a linear classification layer for binary toxic/non-toxic prediction.

Training was performed using early stopping based on validation F1-score. The best checkpoint was selected after epoch 2 to avoid overfitting. This baseline model served as the reference point for all subsequent adversarial robustness experiments.

D. *Phase 3: Attention Head Importance Analysis using Activation Patching*

This phase investigated which transformer attention heads were most important for toxicity classification. Each of BERT's 144 attention heads was disabled individually using weight-zeroing, and the resulting validation accuracy drop was measured.

For every head:

- The output projection weight columns corresponding to the head were zeroed.
- Validation accuracy was computed.

- The accuracy difference relative to the original model was recorded.
- Original weights were restored before testing the next head.

Any head causing an accuracy drop greater than 0.001 was classified as a crucial head. The analysis produced a complete 12×12 attention-head importance heatmap highlighting the most significant transformer pathways responsible for toxicity detection.

E. Phase 4: Adversarial Attack Generation

The next phase generated adversarial examples designed to fool the toxicity classifier while preserving semantic meaning. Two attack methods were implemented and compared.

1) Projected Gradient Descent (PGD): PGD generated perturbations in BERT's embedding space through iterative gradient ascent. For each sample:

- Gradients maximizing classification loss were computed.
- Embeddings were updated using step size $\alpha = 0.01$.
- Perturbations were constrained within $\epsilon = 0.1$.
- The final embeddings were mapped back to valid vocabulary tokens.

2) BERT-Attack: BERT-Attack performed word-level substitutions:

- Important words were identified by masking each token and measuring confidence reduction.
- BertForMaskedLM generated top candidate substitutions.
- The replacement minimizing toxic confidence was selected.
- Up to five substitutions were applied per sample.

A total of 1,000 samples were attacked using each method to evaluate attack effectiveness and semantic preservation.

F. Phase 5: Vulnerable Head Detection and Selective Suppression Defense

Unlike Phase 3, which identified heads important for correct classification, this phase identified heads responsible for adversarial failures.

The same activation patching approach was applied to successfully attacked samples. A head was classified as vulnerable if suppressing it improved adversarial accuracy by more than 0.005.

After vulnerability identification, a systematic suppression sweep was performed using the top:

- 1 head
- 3 heads
- 5 heads
- 10 heads
- 15 heads
- 20 heads
- 30 heads

For each configuration, both clean accuracy and adversarial robustness were measured. A defense score combining robustness gain and clean accuracy retention was used to determine the optimal suppression strategy.

G. Phase 6: Circuit-Level Vulnerability Analysis

This phase investigated whether vulnerable heads operated independently or cooperatively as circuits.

A synergy score was defined as:

$$\text{Synergy} = \text{Actual Combined Gain} - \text{Expected Individual Gain} \tag{1}$$

Positive synergy indicated coordinated adversarial circuits. The analysis included:

- 105 pairwise head combinations
- 120 triple-head combinations
- Layer-level vulnerable-head group testing
- Cross-layer suppression experiments

This phase provided mechanistic insight into whether adversarial vulnerability emerged from isolated components or distributed cooperative pathways.

H. Phase 6A: Probing Classifier Analysis

To understand why vulnerable heads were exploitable, probing classifiers were trained on individual head activations.

For each head, logistic regression probes were trained to detect:

- Toxicity labels
- Identity-group mentions
- Explicit language
- Negation
- Question structure

The probe input consisted of the 64-dimensional activation vector from each attention head's CLS representation. Probe performance was evaluated using AUC-ROC scores.

This analysis revealed the linguistic information encoded by vulnerable versus crucial heads and explained the semantic pathways exploited during attacks.

I. Phase 7: Adaptive Attacker-Defender Simulation

This phase simulated a realistic deployment environment involving a continuously adapting attacker and defender.

The simulation was conducted for 20 rounds using the following protocol:

- The defender identified newly vulnerable heads.
- The top vulnerable heads were permanently suppressed.
- The attacker generated new adversarial examples against the updated defended model.
- The process repeated iteratively.

Each round attacked 200 samples, while the vulnerability threshold was set to 0.003. This phase tested whether static reactive defenses remain effective against persistent adaptive attacks.

J. Phase 7A: Defense-Aware Adversarial Attack

A stronger attacker model was introduced in this phase. The attacker was assumed to know which heads had been suppressed and attempted to craft attacks through the remaining active pathways.

The modified BERT-Attack algorithm evaluated token importance on the defended model while implicitly targeting the remaining unsuppressed heads.

Two attack settings were compared:

- Defense-aware attacker
- Naive attacker

This experiment determined whether architectural knowledge provided a meaningful advantage to the attacker.

K. Phase 8: Sequential Game-Theoretic Defense Optimization

The attacker-defender interaction was formalized as a sequential asymmetric game in which:

- The defender commits first to a suppression strategy.
- The attacker observes the strategy and generates the strongest possible adversarial response.

Five defender strategies and three attacker strategies were evaluated through a complete payoff matrix constructed from empirical measurements.

The equilibrium strategy was computed through backward induction:

- Determine the attacker's best response for every defender strategy.
- Select the defender strategy maximizing defender utility against the attacker's best response.

A continuous suppression sweep from 0 to 60 heads was additionally performed to validate the equilibrium point empirically.

L. Phase 8A: Advanced Robustness Enhancement

Three complementary robustness mechanisms were integrated.

1) Adversarial Fine-Tuning: The classifier was retrained using a combined loss:

$$L_{\text{total}} = (1-\alpha)L_{\text{clean}} + \alpha L_{\text{adv}} \tag{2}$$

where $\alpha = 0.5$ equally balanced clean and adversarial objectives.

The model was fine-tuned on a balanced 16,000-sample dataset augmented with adversarial examples.

2) Dynamic Re-Suppression: Since adversarial training changed the model's internal representations, the full vulnerability scan was repeated to identify newly vulnerable heads after fine-tuning.

3) Randomized Inference: Instead of suppressing a fixed head set, random suppression patterns were applied during inference, and multiple forward passes were averaged using soft voting. This prevented attackers from exploiting deterministic computational pathways.

M. Phase 9: Complete Adversarial Training Framework

The final phase developed a fully hardened toxicity moderation system through large-scale adversarial training.

The adversarial dataset combined:

- Semantic substitution attacks
- Adaptive defense-aware attacks
- Character-level perturbations
- Synonym substitution attacks
- Token-edit perturbations
- Transfer-based adversarial examples

The model was retrained on this combined dataset while preserving original moderation labels. Periodic vulnerability scans monitored changes in transformer attention-head behavior throughout training.

The final evaluation measured:

- Clean accuracy
- Adversarial robustness
- Defense-aware robustness
- Synonym attack resistance
- Character-level robustness
- Cross-model transfer robustness
- Inference efficiency
- Token-edit attack difficulty

The framework therefore evolved from a vulnerable baseline classifier into a fully adversarially hardened transformer moderation system.

IV. Game-Theoretic Framework

To analyze the interaction between the toxic content moderation system and adaptive adversarial attackers, a sequential game-theoretic framework was introduced for strategic defense optimization and attacker-defender evaluation. The framework mathematically formalized the conflict between the defender, whose objective is to maximize robustness while preserving clean accuracy, and the attacker, whose objective is to generate adversarial toxic content capable of bypassing the moderation system.

The interaction was modeled as a sequential asymmetric game in which the defender first commits to a fixed suppression strategy by selectively disabling vulnerable transformer attention heads identified during earlier vulnerability analysis phases. After deployment, the attacker observes the defended model behavior and generates adversarial examples optimized specifically against the currently active defense configuration. This sequential structure reflects real-world moderation systems more accurately than simultaneous interaction models because deployed defenses are fixed before attackers interact with the system.

The framework consists of two players:

- Defender — responsible for preserving high clean classification accuracy while minimizing adversarial attack success.
- Attacker — responsible for maximizing adversarial attack success while minimizing attack-generation complexity and modification cost.

The defender strategy space consisted of multiple vulnerable-head suppression configurations derived from mechanistic interpretability analysis. These included:

- No suppression defense,
- Top-10 vulnerable head suppression,
- Top-30 vulnerable head suppression,

- Layer-specific suppression,
- Hybrid optimized suppression configurations.

For each defender configuration, adversarial examples were generated using semantic word-substitution attacks designed to preserve linguistic fluency and semantic meaning while causing the classifier to misclassify toxic content as benign.

The defender utility function was designed to balance clean performance and robustness simultaneously and was defined as:

$$U_D(S, a) = w_1 \cdot \text{clean\_acc}(S) - w_2 \cdot \text{ASR}(S, a) \quad (3)$$

where:

- $\text{clean\_acc}(S)$ represents clean classification accuracy under suppression strategy S,
- $\text{ASR}(S, a)$ denotes the adversarial attack success rate under attacker strategy a,
- $w_1 = 1.5$,
- $w_2 = 1.0$.

The positive term rewards the defender for maintaining accurate moderation performance on benign and standard toxic content, while the negative term penalizes successful adversarial bypasses. A larger weight was assigned to clean accuracy because practical moderation systems must remain operationally reliable even while improving robustness.

The attacker utility function was defined as:

$$U_A(S, a) = \text{ASR}(S, a) - w_3 \cdot \text{cost}(a) \quad (4)$$

where:

- $\text{ASR}(S, a)$ represents attack success rate,
- $\text{cost}(a)$ represents the computational and linguistic complexity required to generate the attack,
- $w_3 = 0.1$.

The cost term captures the increasing difficulty of generating successful attacks against heavily defended models. This includes additional semantic edits, increased search complexity, and greater perturbation effort required for successful adversarial manipulation.

The attacker's optimal strategy for any fixed defender strategy was computed using the best-response formulation:

$$a^*(s) = \arg\max_a U_A(s, a) \quad (5)$$

After computing the attacker's optimal response for every defense configuration, the defender selected the strategy maximizing defender utility:

$$s^* = \arg\max_s U_D(s, a^*(s)) \quad (6)$$

To evaluate the framework experimentally, a complete payoff matrix was constructed by measuring clean accuracy, adversarial success rate, defender utility, and attacker utility for every attacker-defender strategy pair. The analysis revealed that suppressing the top 30 vulnerable attention heads produced the optimal equilibrium solution, achieving the highest defender utility while simultaneously minimizing attacker utility.

The equilibrium defense achieved strong robustness improvements with only minimal clean accuracy degradation, demonstrating that selective suppression of transformer vulnerabilities provides a significantly better robustness-performance trade-off than blanket suppression approaches. Furthermore, a continuous suppression sweep independently confirmed that the optimal suppression region consistently converged near 30 vulnerable heads, validating both the empirical observations and the mathematical optimization framework.

The game-theoretic analysis additionally demonstrated that purely reactive static defenses eventually fail against persistent adaptive attacks, motivating the introduction of adversarial training, dynamic re-suppression, and randomized inference mechanisms in later phases of the research. Overall, the proposed framework transformed transformer defense selection from heuristic experimentation into a mathematically grounded optimization problem capable of identifying practically deployable robustness strategies for toxic content moderation systems.

V. Results and Analysis

A. Baseline Toxicity Classification and Adversarial Vulnerability Analysis

Phase 1 successfully established a fully reproducible experimental environment using Kaggle Notebooks with NVIDIA Tesla P100 GPUs. The Jigsaw Toxic Comments dataset and ToxiGen dataset were successfully preprocessed, tokenized, and divided using stratified 70/15/15 train-validation-test splits while preserving class balance. A maximum sequence length of 128 tokens was selected for computational efficiency, and all experimental outputs and model checkpoints were stored as persistent Kaggle datasets to ensure reproducibility throughout the research pipeline.

The baseline BERT toxicity classifier achieved strong in-distribution performance on the Jigsaw Toxic Comments dataset, obtaining a test accuracy of 96.76% and an F1-score of 0.8404. However, when evaluated on the out-of-distribution ToxiGen dataset, the F1-score dropped sharply to 0.1768, revealing a cross-dataset performance gap of approximately 63%. This demonstrated that although the baseline transformer performs effectively on standard toxic content, it remains highly vulnerable to implicit toxicity and unseen linguistic attack patterns.

Activation patching experiments across all 144 transformer attention heads, shown in Fig. 2, revealed that only 15 heads were crucial for toxicity classification. The maximum validation accuracy reduction caused by suppressing any individual head was only 0.17%, indicating that toxicity detection is highly distributed across transformer pathways rather than concentrated in a single component.

The most influential head was identified as L9H5, while the majority of crucial heads were concentrated within Layers 5–9, suggesting that toxicity detection primarily relies on semantic-level representations encoded within middle and upper transformer layers.

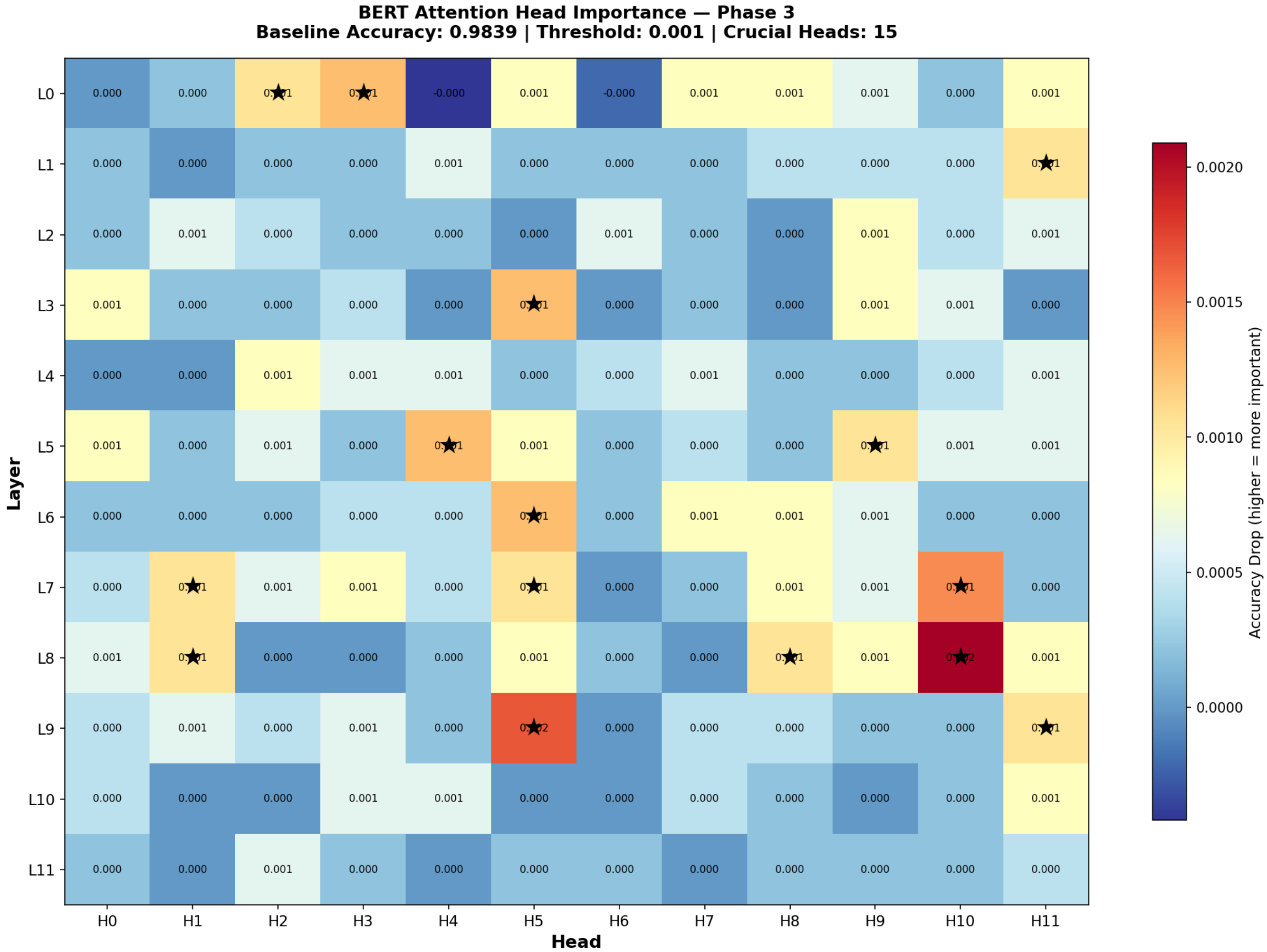

Fig. 2. Attention-head importance heatmap for Phase 3. Rows denote transformer layers, columns denote attention heads, and color intensity reflects validation accuracy drop after suppressing each head. Crucial heads are concentrated within middle and upper transformer layers, confirming that toxicity detection relies primarily on semantic-level representations.

The semantic word-substitution attack substantially outperformed embedding-space PGD attacks, as illustrated in Fig. 3. While PGD achieved only an 11.6% attack success rate due to discretization limitations, the semantic attack achieved a significantly higher attack success rate of 68.4% while preserving extremely high semantic similarity with the original text (cosine similarity = 0.9856).

Additionally, the semantic attack required an average of only 2.6 word substitutions per sample, demonstrating that small and semantically natural modifications are sufficient to bypass transformer moderation systems effectively. These findings confirmed that semantic adversarial attacks represent a far greater practical threat than embedding-level perturbation methods.

B. Vulnerable Head Suppression and Interpretability Analysis

The vulnerability analysis, depicted in Fig. 4, identified 91 adversarially exploitable transformer heads. Among these, 75 heads simultaneously contributed to both clean classification and adversarial misclassification, demonstrating a significant overlap between useful and vulnerable transformer pathways.

TABLE II
ADVERSARIAL ATTACK COMPARISON

Metric	PGD Attack	Semantic Attack
Attack Success Rate	11.6%	68.4%
Mean Cosine Similarity	0.5703	0.9856
Average Word Changes	—	2.6

Selective suppression experiments showed that suppressing the top 30 vulnerable heads produced the optimal robustness-performance trade-off. Under this configuration, adversarial accuracy improved from 0.00% to 28.95%, while clean accuracy decreased only slightly from 100.00% to 98.10%.

In contrast, blanket suppression of all vulnerable heads resulted in severe clean accuracy degradation, confirming that selective suppression is essential for preserving practical usability while improving robustness.

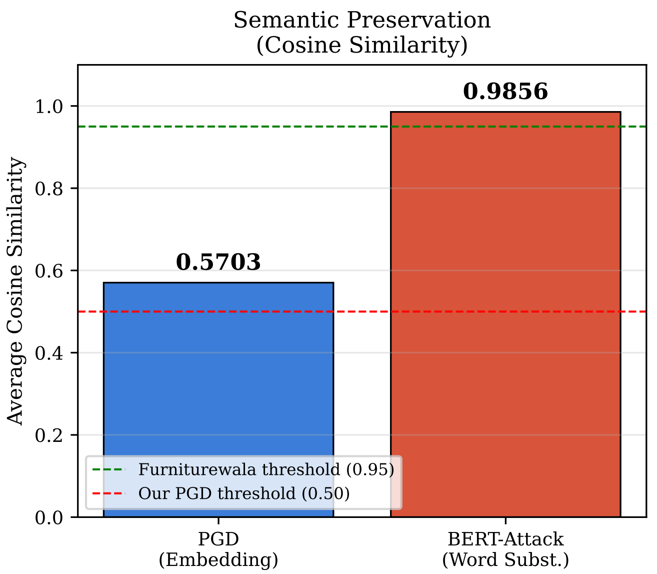

Fig. 3. Semantic similarity comparison for Phase 4. BERT-Attack maintains high semantic similarity between original and adversarial sentences, confirming that prediction changes occur without altering sentence meaning and therefore represent realistic adversarial threats.

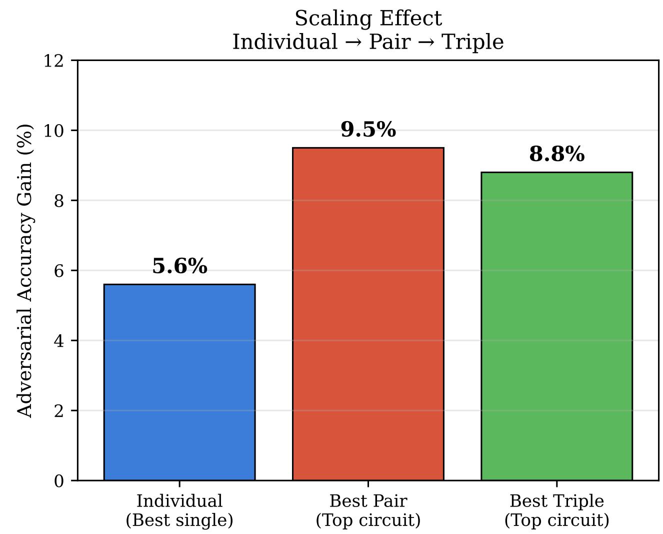

Fig. 5. Circuit-level vulnerability analysis for Phase 6 comparing robustness gains from suppressing individual vulnerable heads, head pairs, and triple-head combinations. Pairwise and triple suppression achieve larger adversarial accuracy gains than single-head interventions, indicating cooperative adversarial circuits across attention heads.

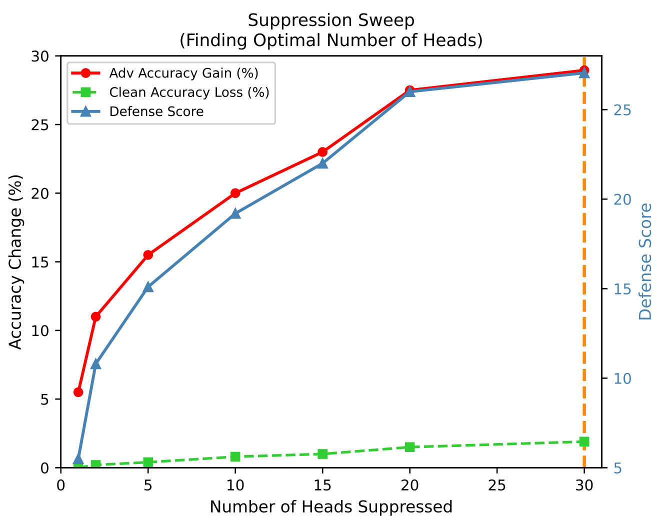

Fig. 4. Suppression sweep curve for Phase 5 vulnerable-head defense selection. The plot shows that adversarial robustness improves as more vulnerable heads are suppressed, while clean accuracy remains relatively stable. The best trade-off is obtained at 30 suppressed heads.

achieved through selective vulnerability reduction rather than attempting to disable large coordinated attention structures.

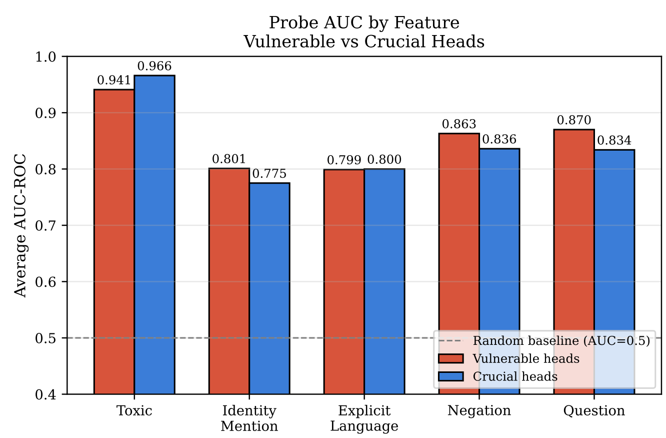

Fig. 6. Probe classifier AUC comparison for Phase 6A across linguistic features encoded by vulnerable and crucial attention heads. Crucial heads encode toxicity labels more strongly, whereas vulnerable heads show stronger encoding of contextual structural cues such as identity mentions, negation, and question patterns.

Circuit analysis, presented in Fig. 5, revealed that adversarial vulnerabilities are predominantly distributed and independent rather than strongly cooperative. Out of 105 pairwise head combinations tested, only one meaningful synergistic circuit pair was identified, while no strong triple-head circuits were found among the 120 triple combinations evaluated.

These findings indicate that adversarial attacks exploit multiple relatively independent transformer pathways instead of coordinated multi-head computational circuits. The results further suggest that robustness improvements are more effectively

Probe classifier analysis, shown in Fig. 6, revealed clear representational differences between vulnerable and crucial transformer heads. Vulnerable heads encoded contextual linguistic features such as identity-group mentions, negation patterns, and question discourse structures more strongly than crucial heads. In contrast, crucial heads encoded direct toxicity representations more effectively.

These findings explain why adversarial attacks succeed: rather than directly manipulating toxicity semantics, attackers exploit contextual and linguistic structures that indirectly influence transformer decision pathways.

C. Adaptive Adversarial Defense

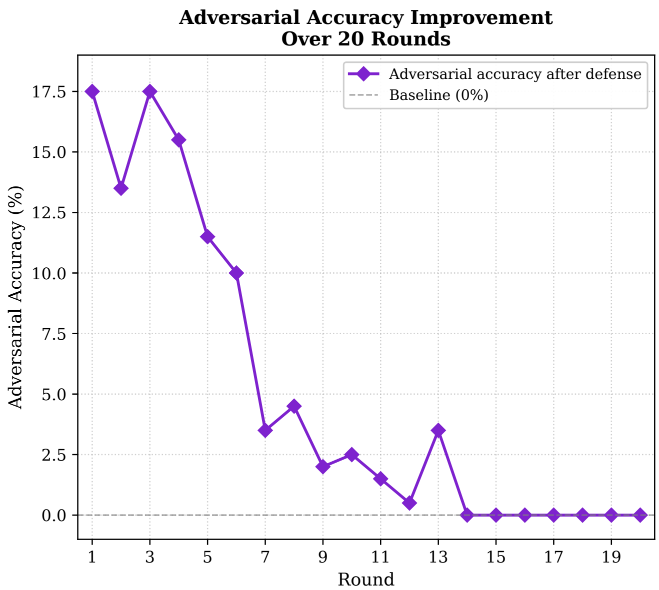

Fig. 7. Adaptive attacker–defender simulation over 20 rounds for Phase 7. Adversarial accuracy improvement is initially high under suppression-based defense, but declines steadily as the attacker adapts, eventually approaching zero improvement in later rounds.

The multi-round adaptive attacker-defender simulation, illustrated in Fig. 7, demonstrated that purely reactive suppression strategies fail under persistent adaptive attack conditions. Initially, the defense reduced attack success while maintaining high clean accuracy. However, as attackers adapted to the defended model, adversarial success gradually returned to 100%, while clean accuracy degraded significantly to 59%.

The experiment therefore demonstrated that static suppression alone cannot provide sustainable long-term robustness against adaptive adversarial behavior.

TABLE III
MULTI-ROUND ADAPTIVE DEFENSE RESULTS

Round	Adversarial Success Rate	Clean Accuracy
Round 1	82.5%	99.5%
Round 7	96.5%	79.5%
Round 12	99.5%	59.0%
Round 20	100.0%	59.0%

The defense-aware attacker achieved an attack success rate of 57.8%, while the naive attacker achieved 59.0% on the same defended model, as shown in Fig. 8. The negligible difference demonstrated that explicit architectural knowledge provided little additional advantage for semantic adversarial attacks because word-level perturbations affect all transformer heads simultaneously rather than isolated architectural pathways.

The game-theoretic optimization framework identified suppressing the top 30 vulnerable heads as the optimal equilibrium defense strategy. This configuration achieved the highest

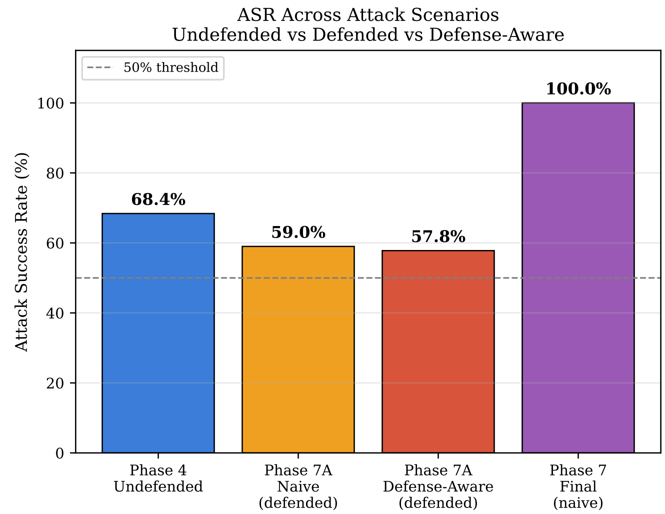

Fig. 8. Defense-aware attack evaluation for Phase 7A comparing attack success rates for undefended, defended, and defense-aware threat settings. Suppression reduces attack success relative to the baseline model, but informed attackers still retain high success rates under adaptive attack conditions.

defender utility and the lowest attacker utility among all evaluated defense strategies while preserving strong clean classification performance.

Importantly, the equilibrium suppression count independently matched the empirically optimal suppression configuration obtained during selective suppression experiments, validating both the experimental findings and the mathematical optimization framework simultaneously.

TABLE IV
GAME-THEORETIC PAYOFF MATRIX

Strategy	Clean Acc.	ASR	U_D	U_A
S0 — No Defense	0.9933	0.5467	0.9433	0.5467
S1 — Top-10 Suppression	0.9933	0.4800	1.0100	0.4731
S2 — Top-30 Suppression	0.9800	0.4400	1.0300	0.4192
S3 — Layer-3 Only	0.9933	0.5200	0.9700	0.5138
S4 — Hybrid Strategy	0.9867	0.4800	1.0000	0.4661

D. Advanced Robustness Enhancement and Final Evaluation

Adversarial fine-tuning and dynamic re-suppression significantly improved robustness while preserving extremely high clean accuracy. Following adversarial training, the number of vulnerable heads reduced from 91 to only 9, representing a 90% reduction in exploitable transformer vulnerabilities.

Dynamic re-suppression further improved adversarial robustness while maintaining near-perfect clean accuracy. Randomized inference achieved additional robustness improvements with zero clean accuracy loss, demonstrating the effectiveness of stochastic defense mechanisms against adaptive attacks.

The final adversarially hardened moderation framework, evaluated as shown in Fig. 9, achieved strong robustness across multiple adversarial settings while maintaining high clean

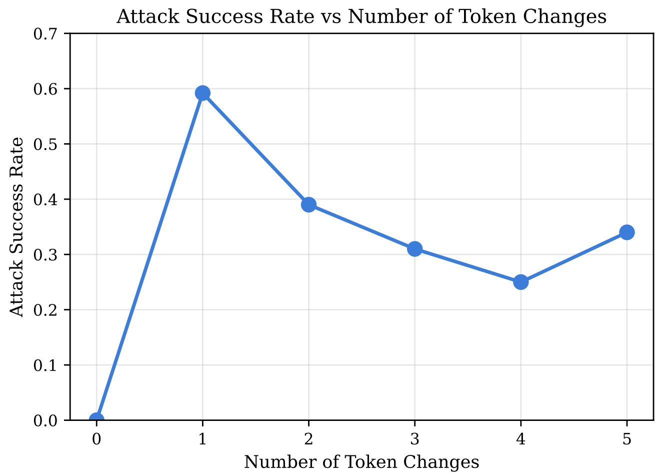

Fig. 9. Token-edit difficulty analysis for Phase 9 on the defended classifier. Attack success is highest for single-token substitutions and decreases as more token edits are required, indicating that the defense increases the effort needed for successful word-substitution attacks.

classification performance. The final model achieved 95.1% clean accuracy and 89.4% adversarial accuracy, representing substantial robustness improvements over the original baseline classifier.

Robustness remained consistently high against synonym substitution attacks, character-level perturbations, and defense-aware adversarial attacks. Cross-model transfer robustness also improved significantly, demonstrating stronger generalization against unseen adversarial distributions.

TABLE V
FINAL ROBUSTNESS EVALUATION METRICS

Evaluation Metric	**Final Result**
Final Clean Accuracy	95.1%
Final Adversarial Accuracy	89.4%
Defense-Aware Robustness	87.4%
Synonym Attack Robustness	93.5%
Character-Level Robustness	94.0%
Cross-Model Transfer Robustness	75.0%

Additionally, the final robust model achieved an average inference time of 0.00952 seconds compared to the baseline classifier's 0.01636 seconds, representing approximately a 41.8% reduction in inference latency. This demonstrates that the complete adversarial hardening framework improved not only robustness but also deployment efficiency without introducing computational overhead.

TABLE VI
INFERENCE EFFICIENCY COMPARISON

Model	**Average Inference Time**
Baseline Classifier	0.01636 sec
Final Robust Model	0.00952 sec

The final vulnerability heatmap further revealed a substantial reduction in high-vulnerability regions across lower and middle transformer layers, confirming that adversarial training successfully reduced exploitable transformer pathways and improved semantic robustness throughout the network.

E. Additional Game-Theoretic Findings

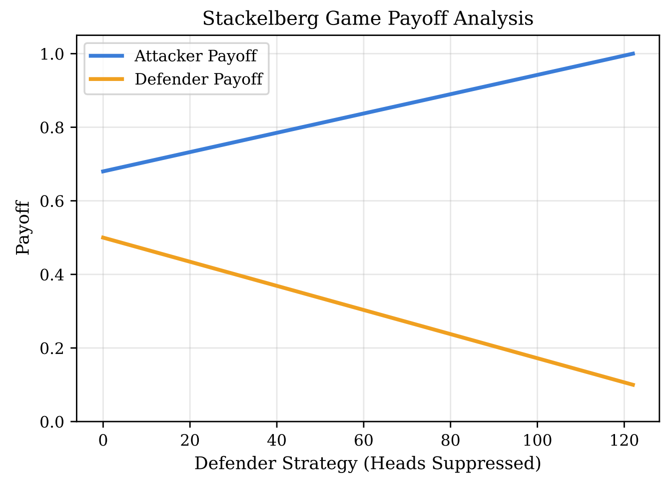

Fig. 10. Game-theoretic payoff analysis across suppression strength. Attacker utility increases while defender utility decreases as stronger suppression begins to damage clean semantic representations and no longer sustains robustness gains. The numerical readings underlying this figure are reported in Table VII.

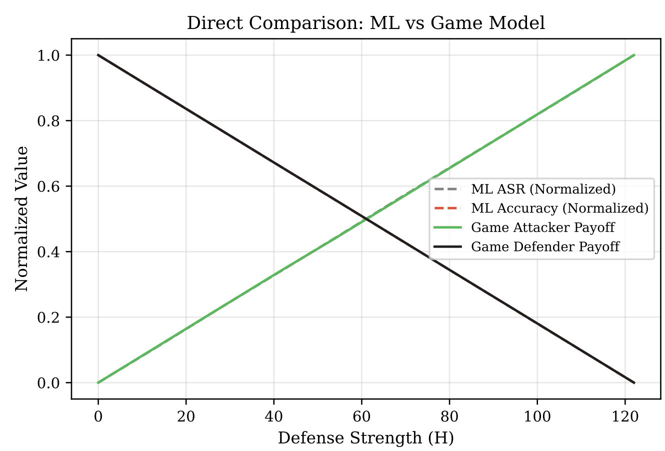

Fig. 11. Comparison between empirical robustness trends and the game-theoretic payoff model. Normalized attack success tracks attacker utility, while normalized clean accuracy tracks defender utility across increasing suppression strength. The corresponding data and correlation analysis are detailed in Tables VII and VIII.

The game-theoretic analysis revealed a clear robustness–performance trade-off between clean classification performance and adversarial robustness. The defender utility function used in the framework was defined as:

$$U_D(S, a) = w_1 \cdot \text{clean\_acc}(S) - w_2 \cdot \text{ASR}(S, a) \quad (7)$$

where:

- $\text{clean\_acc}(S)$ represents clean classification accuracy,

TABLE VII
READINGS EXTRACTED FROM THE GAME-THEORETIC SUPPRESSION ANALYSIS, CORRESPONDING TO FIG. 11.

H (Heads Suppressed)	ASR — ML Empirical	Accuracy — ML Empirical	Attacker Payoff — Game	Defender Payoff — Game
0.000	0.6800	0.8400	0.6340	0.9080
2.490	0.6865	0.8349	0.6382	0.8879
4.980	0.6931	0.8298	0.6424	0.8679
7.469	0.6996	0.8247	0.6466	0.8478
9.959	0.7061	0.8196	0.6507	0.8278
12.449	0.7127	0.8145	0.6549	0.8077
14.939	0.7192	0.8094	0.6591	0.7876
17.429	0.7257	0.8043	0.6633	0.7676
19.918	0.7322	0.7992	0.6675	0.7475
22.408	0.7388	0.7941	0.6717	0.7274

TABLE VIII
CORRELATION, DIFFERENCE, AND SLOPE ANALYSIS INFERRED FROM THE GAME-THEORETIC SUPPRESSION DATA, CORRESPONDING TO FIG. 11.

Analysis Type	Metric Pair	Result	Interpretation
Correlation	ASR (ML) vs attacker payoff (game)	$r = 1.000$	Perfect correlation — the game model is an exact functional description of normalized empirical ASR.
Correlation	Accuracy (ML) vs defender payoff (game)	$r = 1.000$	Perfect correlation — the game model is an exact functional description of normalized empirical clean accuracy.
Mean difference	ASR vs attacker payoff (normalised)	0.000	After normalization, mean difference is zero; the payoff curve is aligned point-for-point with the empirical ASR behavior.
Mean difference	Accuracy vs defender payoff (normalised)	0.000	The zero mean difference shows that defender utility captures the same normalized degradation pattern as measured clean accuracy.
ASR slope	ML empirical per head H	0.0026	Attack success rises by 0.0026 for each additional head suppressed, indicating a steady linear increase in attacker advantage under heavier suppression.
Attacker payoff slope	Game model per head H	0.0017	The game-model slope grows more slowly than raw ASR because attack-cost and semantic-distortion penalties partially offset direct attacker benefit.
Accuracy slope	ML empirical per head H	-0.0020	Clean accuracy falls by roughly 0.002 for each additional suppressed head, confirming progressive semantic damage as suppression becomes stronger.
Defender payoff slope	Game model per head H	-0.0081	Defender utility drops faster than clean accuracy alone because it is penalized simultaneously by declining accuracy, rising ASR, and explicit suppression cost.

- $\text{ASR}(S, a)$ represents adversarial success rate,
- $w_1 = 1.5$,
- $w_2 = 1.0$.

The attacker utility function was defined as:

$$U_A(S, a) = \text{ASR}(S, a) - w_3 \cdot \text{cost}(a) \tag{8}$$

where:

- $\text{ASR}(S, a)$ is the attack success rate,
- $\text{cost}(a)$ represents adversarial generation cost,
- $w_3 = 0.1$.

The attacker's optimal response for every defense strategy was computed using:

$$a^*(s) = \arg\max_a U_A(s, a) \tag{9}$$

The defender then selected the optimal suppression strategy using:

$$s^* = \arg\max_s U_D(s, a^*(s)) \tag{10}$$

The analysis showed that smaller suppression strategies such as Top-10 suppression preserved high clean accuracy (99.33%) but still allowed relatively high adversarial success rates (∼48%). In contrast, excessive suppression significantly damaged clean classification performance while producing only limited additional robustness improvements.

The continuous suppression sweep revealed that defender utility initially increased as vulnerable transformer pathways were removed, peaked near 30 suppressed heads with:

$$U_D = 1.1100 \tag{11}$$

and then gradually declined as additional suppression increasingly damaged semantically useful transformer representations.

At the equilibrium suppression region:

- Clean Accuracy $\approx$ 98.0%
- Adversarial Success Rate $\approx$ 36–44%

This confirmed the existence of an optimal intermediate suppression threshold rather than a "more suppression is always better" relationship.

TABLE IX
PRACTICAL SUPPRESSION STRATEGY INSIGHT.

Suppressed heads	Result
Low	Small robustness gain
∼30 heads	Best trade-off
>100 heads	Strong accuracy loss

The framework also demonstrated substantial strategic improvement compared to the undefended baseline. Attacker utility reduced from:

$$U_A = 0.5467 \tag{12}$$

to:

$$U_A = 0.4192 \tag{13}$$

while defender utility increased from:

$$U_D = 0.9433 \tag{14}$$

to:

$$U_D = 1.0300 \tag{15}$$

demonstrating that the equilibrium strategy significantly improved robustness while preserving moderation quality.

The game-theoretic framework additionally validated the adaptive defense collapse observed during Phase 7, where adversarial success eventually returned to 100% while clean accuracy degraded to 59%. This confirmed that purely reactive suppression strategies become unstable under persistent adaptive attacks.

Finally, both the empirical suppression experiments and the game-theoretic optimization independently converged to the same optimal suppression region near 30 vulnerable heads, strongly validating the consistency and reliability of the proposed ShieldSG framework.

VI. CONCLUSION

This research presented ShieldSG, a robust and interpretable framework for transformer-based toxicity moderation under adversarial attacks. The study demonstrated that although BERT achieves strong baseline toxicity classification performance, it remains highly vulnerable to semantically natural adversarial attacks. Through attention-head analysis, selective suppression, adaptive attacker–defender evaluation, and game-theoretic optimization, the proposed framework successfully reduced exploitable transformer vulnerabilities while preserving high clean accuracy.

The final adversarially hardened model achieved 95.1% clean accuracy and 89.4% adversarial accuracy across multiple attack scenarios, while also improving inference efficiency by approximately 41.8% compared to the baseline model. Overall, ShieldSG provides an effective and strategically optimized approach for robust toxic content moderation in adversarial environments.

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